

# Right Haar measure on the affine group refutes the Modular Collapse Conjecture

Autonomous mathematical discovery run `fa_banach_001`

27 August 2026

**Status.** Candidate full negative answer, likely valid pending expert review.

## 1 Source conjecture

Inoué defines a left quasi-invariant measure on a locally compact topological quasigroup  $Q$  to be a nonzero regular Borel measure  $\mu$  for which

$$(L_a)_*\mu = j(a)\mu \quad (a \in Q)$$

with positive scalar cocycle  $j$ . Conjecture 1, the *Modular Collapse Conjecture*, asserts that if  $Q$  satisfies

$$((xy)z)y = x(y(z y)) \tag{N1}$$

and admits such a measure, then  $j \equiv 1$  [1, Conjecture 1, PDF p. 16].

We do not attempt to solve this existence problem here. Instead, the present paper isolates the cocycle mechanism that would become relevant once such a measure is available.

## 11 Towards a Measure-Theoretic Interpretation of Kunen's Theorem

The purpose of this paper is to develop a framework in which Haar-type structures on topological quasigroups can be studied via quasi-invariant measures and modular cocycles. From this measure-theoretic viewpoint, loop identities may be interpreted as a form of unimodularity in the translation geometry of quasigroups.

This perspective naturally leads to the following conjecture.

**Conjecture 1** (Modular Collapse Conjecture). *Let  $Q$  be a locally compact Hausdorff topological quasigroup satisfying the Moufang-type identity (N1). Assume that  $Q$  admits a quasi-invariant measure with modular cocycle  $j$ . Then the cocycle collapses:*

$$j \equiv 1.$$

*In particular, the translation geometry becomes unimodular in the measure-theoretic sense.*

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We give a counterexample which satisfies the stronger, two-sided quasi-invariance hypothesis used elsewhere in the source.

## 2 Counterexample

**Theorem 1.** *The Modular Collapse Conjecture is false. In fact, there is a connected two-dimensional Lie group  $G$ , regarded as a topological quasigroup, and a nonzero regular Borel measure  $\nu$  such that*

$$(L_g)_*\nu = j(g)\nu, \quad (R_g)_*\nu = \nu \quad (g \in G),$$

but  $j \not\equiv 1$ .

*Proof.* Let

$$G = \{(a, b) : a > 0, b \in \mathbb{R}\}, \quad (a, b)(a', b') = (aa', b + ab').$$

This is the group of orientation-preserving affine transformations of the line. It is a locally compact Hausdorff topological group and hence a topological quasigroup. Associativity gives, for every  $x, y, z \in G$ ,

$$((xy)z)y = xyz y = x(y(zy)),$$

so (N1) holds.

On the open half-plane  $G \subset \mathbb{R}^2$ , define

$$d\nu(a, b) = \frac{da db}{a}.$$

The density is positive and continuous, so  $\nu$  is a nonzero regular Borel measure, finite on compact sets.

Fix  $g = (\alpha, \beta) \in G$ . Left translation is

$$L_g(a, b) = (\alpha a, \beta + \alpha b).$$

With

$$u = \alpha a, \quad v = \beta + \alpha b,$$

we have  $du dv = \alpha^2 da db$  and  $a = u/\alpha$ . Therefore, for every nonnegative Borel function  $f$ ,

$$\begin{aligned} \int_G f d((L_g)_*\nu) &= \int_G f(\alpha a, \beta + \alpha b) \frac{da db}{a} \\ &= \frac{1}{\alpha} \int_G f(u, v) \frac{du dv}{u}. \end{aligned}$$

Thus

$$(L_{(\alpha, \beta)})_*\nu = \alpha^{-1}\nu, \quad j(\alpha, \beta) = \alpha^{-1}.$$

For completeness, right translation is

$$R_g(a, b) = (a\alpha, b + a\beta).$$

Now  $u = a\alpha$ ,  $v = b + a\beta$ , so  $du dv = \alpha da db$  and again  $a = u/\alpha$ . Hence

$$\begin{aligned} \int_G f d((R_g)_*\nu) &= \int_G f(a\alpha, b + a\beta) \frac{da db}{a} \\ &= \int_G f(u, v) \frac{du dv}{u}. \end{aligned}$$

Consequently  $(R_g)_*\nu = \nu$ : the chosen measure is right Haar and is two-sided quasi-invariant in the source's scalar sense.

All hypotheses of Conjecture 1 hold, while

$$j(2, 0) = \frac{1}{2} \neq 1.$$

This refutes the conjecture. □

### 3 Structural diagnosis

The source correctly derives multiplicativity of  $j$  under  $(N1)$  and two-sided quasi-invariance. The example above obeys that conclusion:

$$j((a, b)(a', b')) = (aa')^{-1} = j(a, b)j(a', b').$$

What fails is the extra step from multiplicativity to triviality. Loop or group structure normalizes  $j$  at the identity but does not eliminate nontrivial positive characters. Indeed, the source explicitly notes this possibility before stating the conjecture.

The counterexample is not sensitive to the one-sided wording of the conjecture:  $\nu$  is invariant under every right translation. The same mechanism works for any non-unimodular locally compact group after choosing right rather than left Haar measure.

### 4 Scope, verification, and novelty

The theorem gives a full negative answer to Conjecture 1 as written. It does not classify additional hypotheses that force modular collapse. Requiring the chosen measure itself to be left invariant forces  $j = 1$  by definition; finding non-tautological rigidity assumptions is a separate question.

The accompanying exact SymPy script verifies both Jacobian-density factors, the identity  $(N1)$  in affine coordinates, and multiplicativity of  $j$ . It is a regression check and is not needed for the proof.

A bounded novelty search on 27 August 2026 covered the four run indexes, the current arXiv record, the exact conjecture name, the paper title, and variants combining the affine group, right Haar measure, and the modular cocycle. No later answer was found. The affine-group Haar formulas are standard; the claimed contribution is their direct application to the new conjecture. Novelty confidence is moderate pending specialist review.

**Human review recommendation.** Check the source’s push-forward convention and the two changes of variables. Also confirm that “a quasi-invariant measure” in Conjecture 1 has the literal meaning of the paper’s definition rather than an unstated left-Haar normalization.

### References

- [1] T. Inoué, *Haar-Type Measures on Topological Quasigroups and Kunen’s Theorem*, arXiv:2603.06174v2 (2026).